Predicting Sales from Campaign Data: Approach Document

Case Study Overview

The objective of this case study is to build a **predictive model** that can estimate product sales generated from influencer marketing campaigns. You will work with real-world, messy campaign data and try to understand which campaign attributes actually drive sales.

You should approach this case as a **Data Scientist working closely with a marketing and growth team**. Your role is to clean unreliable data, explore relationships between campaign variables, build a robust regression model, and translate model results into clear business insights.

This document provides **hints** on how to approach each part of the case study. It does not provide answers, exact steps, or model configurations. All implementation choices are intentionally left to you.

Dataset Overview

You are provided with historical data from multiple influencer marketing campaigns. The dataset reflects a realistic scenario where data has been collected from different sources and is not analysis-ready.

Key Data Fields to Explore

Influencer Reach & Engagement

- Number of followers at the time of the campaign
- Engagement rate on influencer posts

Campaign Spend & Content

- Total ad spend for the campaign
- Content quality score (subjective rating)

Outcome Variable

- Number of product units sold due to the campaign

Additional Information

- Campaign end date (timestamp)
- Notes or free-text fields

Understanding how these fields relate to one another is critical before moving into modeling.

Section A: Data Understanding & Cleaning

Goal

Prepare the raw campaign data so it can be reliably used for analysis and modeling.

Hints

- Start by understanding the scale of the dataset and the range of values in each column.
- Look carefully for formatting issues such as currency symbols, percentage signs, or mixed units.
- Identify missing values and think about whether they represent data collection gaps or realistic campaign scenarios.
- Watch for extreme or unrealistic values (for example, unusually high follower counts or negative spend values) and consider how they may affect the model.

The outcome of this section should be a **clean and consistent dataset** that reflects realistic marketing data.

Section B: Exploratory Data Analysis (EDA)

Goal

Explore the data to understand how different campaign factors relate to sales.

Hints

- Examine relationships between individual campaign features and sales rather than relying on a single metric.
- Compare high-performing and low-performing campaigns to identify noticeable differences.
- Look for patterns, trends, or diminishing returns as campaign spend or influencer reach increases.

- Pay attention to outliers, as they may reveal either data quality issues or important business insights.

Insights

- Focus on insights that can help marketing teams make better decisions about influencer selection, budget allocation, or content strategy.
- Aim to communicate findings in simple terms that non-technical stakeholders can understand.

Section C: Feature Engineering

Goal

Create better input features that improve the predictive power of your model.

Hints

- Think about whether raw values are the best representation of campaign impact or if transformations could make relationships clearer.
- Consider interactions between variables (for example, how engagement and followers together may influence sales).
- Be mindful of feature scale differences and how they may affect model behavior.

Feature engineering decisions should be driven by both **data patterns and marketing intuition**.

Section D: Model Building

Goal

Train a regression model that can predict sales from campaign attributes.

Hints

- Start with a simple baseline model before exploring more complex options.
- Ensure your model can handle noisy and imperfect real-world data.
- Use appropriate validation strategies to avoid overfitting.

The focus should be on building a model that generalizes well, not just one that performs well on training data.

Section E: Model Evaluation & Tuning

Goal

Evaluate how well your model performs and improve its reliability.

Hints

- Use error-based metrics to understand how far predictions are from actual sales values.
- Compare model performance across different configurations to identify meaningful improvements.
- Pay attention to consistency of results across validation folds rather than a single score.

This section is about **model robustness**, not just maximizing performance numbers.

Section F: Business Insights & Pipeline Design

Goal

Translate model results into actionable insights and design a solution that can be reused.

Hints

- Identify which features have the strongest influence on predicted sales and explain why this makes sense from a marketing perspective.
- Think about how this model could be used to evaluate future influencer campaigns before money is spent.
- Design a simple and reliable pipeline that can accept new, messy campaign data and produce predictions.

Success Measurement

- Focus on both predictive accuracy and business usefulness.
- Explain how this solution helps marketing teams make better, data-driven decisions.

Final Notes

- There is no single correct model or feature set for this case study.
- Clear reasoning, structured analysis, and strong justification matter more than complex techniques.
- Treat this exercise as a real-world simulation of how predictive models are built and used in marketing analytics.